

Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology



Department of Mechatronics Engineering

Course No.: MTE 3106

Course Title: Linear Integrated Circuits and Digital Systems Sessional

Experiment No.: 06

Experiment Name: Implementation of Basic Logic Gates Using Integrated Circuits (ICs) on Breadboard.

Remarks

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Date of Experiment:

Date of Submission:

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Experiment Name: Implementation of Basic Logic Gates Using Integrated Circuits (ICs) on Breadboard.

Objectives:

- To comprehend the operational principles and truth tables of fundamental digital logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR) through practical hardware implementation.
- To implement, verify, and analyze the circuit behavior of these basic logic gates utilizing standard TTL ICs on a prototyping breadboard.
- To cultivate essential hands-on skills in digital circuit wiring, component integration, and the troubleshooting of connectivity issues.
- To bridge the theoretical-practical divide by directly correlating theoretical Boolean logic design with tangible, physical outputs.

Introduction:

Digital electronics form the bedrock of modern computing, communication, and control architectures. At the core of these systems are logic gates, which are the fundamental building blocks that execute basic Boolean operations required for sophisticated electronic design. To effectively bridge theoretical logic concepts with real-world applications, practical circuit implementation is strictly necessary.

This experiment investigates the operation of the primary logic gates (7408 (AND), 7432 (OR), 7404 (NOT), 7400 (NAND), 7402 (NOR), 7486(XOR), 74266(XNOR)) using 74-series Transistor-Transistor Logic (TTL) integrated circuits. The TTL ICs utilized in this lab encapsulate multiple independent logic gates within a single package. By physically wiring these components on a breadboard, applying +5V DC regulated power, and monitoring output states via LED indicators, theoretical truth tables are empirically verified. This hands-on methodology not only demonstrates the functional realities of digital logic but also establishes a foundational skill set for designing more complex architectures like arithmetic logic units (ALUs), multiplexers, and memory registers.

In this laboratory experiment, these logic gates are implemented and tested on a breadboard, allowing circuit construction without permanent soldering. This hands-on approach helps students understand the practical operation of logic gates and their role in building larger digital systems such as multiplexers, adders, and flip-flops. The output of each gate is displayed using an LED connected through a current-limiting resistor, while a regulated +5 V DC supply provides the operating voltage required for TTL devices. Jumper wires are used to establish the necessary circuit connections between the ICs, power source, and ground. Through the successful implementation and testing of these logic gates, students gain practical experience in circuit assembly, verification, and troubleshooting. These skills form an essential foundation for advanced studies in digital electronics, microprocessor systems, and embedded system design.

Required Apparatus

1. Breadboard
2. +5V DC Power Supply
3. Digital TTL Logic ICs: IC 7408 (Quad 2-Input AND Gate), IC 7432 (Quad 2-Input OR Gate), IC 7404 (Hex Inverter/NOT Gate), IC 7400 (Quad 2-Input NAND Gate), IC 7402 (Quad 2-Input NOR Gate), IC 7486 (Quad 2-Input XOR Gate), IC 74266 / IC 4077 (Quad 2-Input XNOR Gate)
4. Light Emitting Diodes (LEDs) (7 pcs)
5. Current-Limiting Resistors
6. Input Push Buttons (2 pcs)
7. Jumper Wires (As required)

Working Procedure

1. Preparation: All necessary hardware, including the breadboard, specified TTL ICs, resistors, LEDs, jumper wires, and the +5V DC power supply, was assembled and inspected.
2. Component Mounting: The selected logic ICs were firmly mounted across the central divider of the breadboard to ensure electrical isolation between the opposing rows of pins.
3. Power Distribution (Biasing): To activate the logic chips, the VCC and GND terminals of each IC were wired directly to the +5V rail and the common ground rail, respectively, establishing the necessary operating voltage.
4. Input Configuration: Push-button switches were integrated into the circuit and connected to the gate inputs, serving as manual generators to toggle logic states between HIGH and LOW.
5. Output Monitoring: LEDs, placed in series with $220\sim\Omega$ current-limiting resistors, were connected to the respective output pins of the logic gates to safely visualize the signal status.
6. Testing and Simulation: The input switches were methodically toggled to simulate all possible binary combinations outlined in the standard truth tables.
7. Data Collection and Verification: The resulting output responses for the AND, OR, NOT, NAND, NOR, XOR, and XNOR gates were observed via the LEDs, documented, and cross-referenced against theoretical expectations to verify accurate gate functionality.
8. Teardown: Following successful verification, the power supply was isolated, and the breadboard circuit was carefully dismantled.

Circuit Development

1. AND Gate (IC 7408): The AND gate executes the Boolean multiplication operation ($Y = A \cdot B$). It dictates that the output will register as HIGH (1) strictly when all individual inputs are simultaneously HIGH. Conversely, if any single input is LOW (0), the output is driven LOW. This logic is widely applied in decision-making circuits where multiple conditions must be met. The IC 7408 serves this function, providing four independent 2-input AND gates in one package.

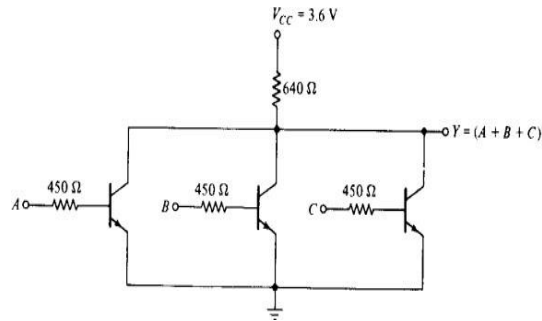
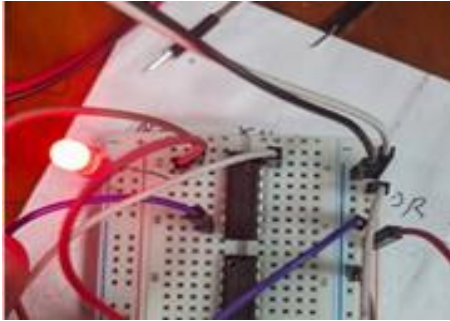


Figure 6.1: Basic NOR Gate

2. OR Gate (IC 7432): The OR gate performs Boolean addition ($Y = A + B$). It yields a HIGH (1) output if at least one of its inputs is actively HIGH. The output drops to LOW (0) only when every input is LOW. This behavior is utilized in systems where independent conditions can individually trigger a shared output response. The IC 7432 is standard for this operation, containing four 2-input OR gates.

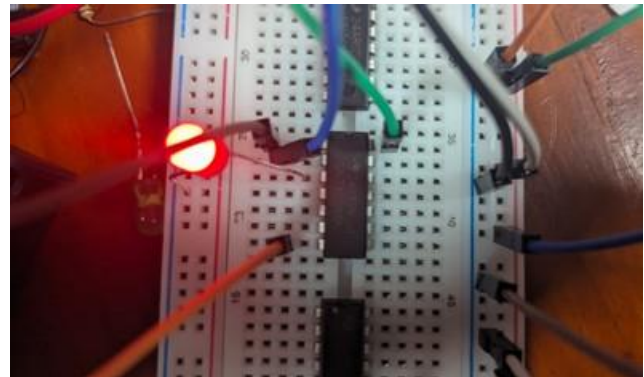
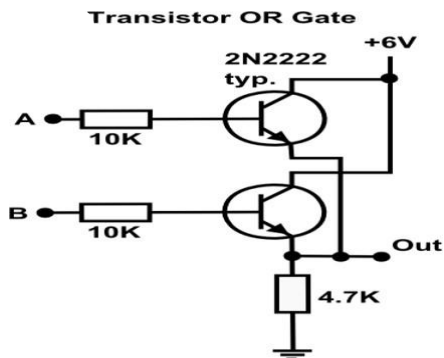


Fig 2: OR Gate

3. NOT Gate (IC 7404): Commonly termed an inverter, the NOT gate performs logical complementation. Featuring a single input and single output, it simply reverses the logic state: a HIGH input yields a LOW output, and a LOW input yields a HIGH output. It is heavily relied upon for signal inversion and logic control. The 7404 IC provides six discrete NOT gates.

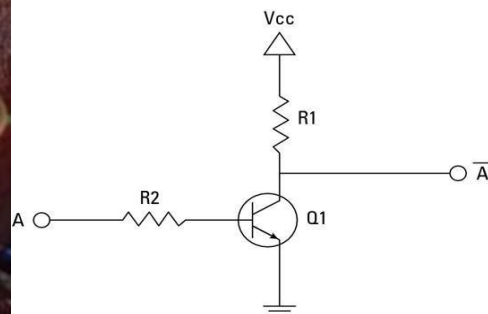
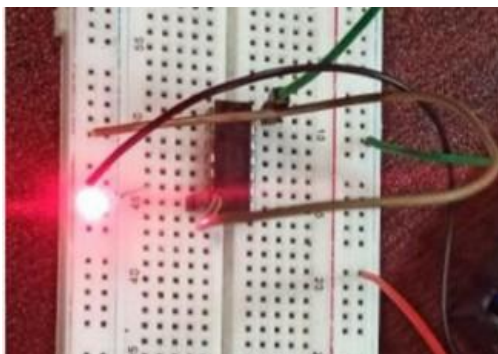


Fig 03: NOT Gate

4. NAND Gate (IC 7400): The NAND (NOT-AND) gate is a universal logic element that produces a LOW (0) output exclusively when all inputs are HIGH (1); otherwise, it defaults to a HIGH output. Functionally, it operates as an AND gate immediately followed by an inverter. Its "universal" designation stems from the fact that any digital logic circuit can be synthesized using NAND gates alone. The IC 7400 contains four independent 2-input NAND gates.

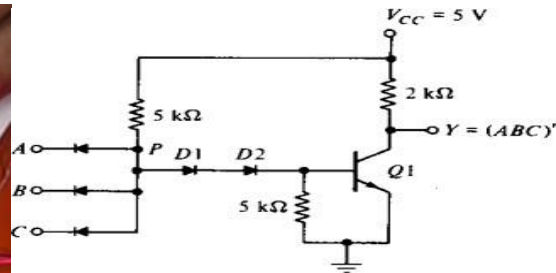
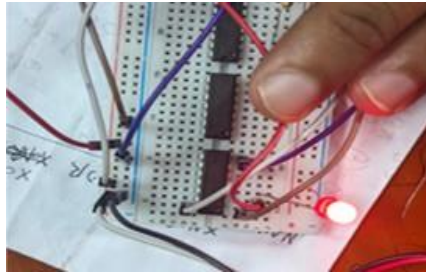


Fig 4: NAND Gate

5. NOR Gate (IC 7402): The NOR (NOT-OR) gate is the second universal logic gate, outputting a HIGH signal solely when all inputs are held LOW. Any presence of a HIGH input immediately forces the output LOW. Like the NAND gate, it can be configured to replicate the function of any other digital gate. The IC 7402 facilitates this with four independent 2-input NOR gates.

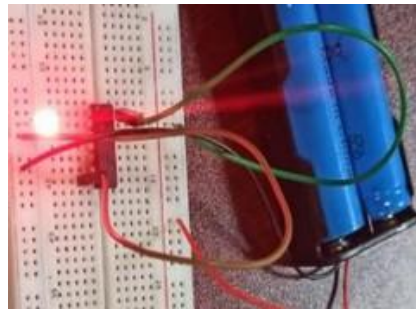


Fig 5: NOR Gate

6. XOR Gate (IC 7486): The Exclusive-OR (XOR) gate serves as a digital inequality detector. It generates a HIGH (1) output only when its inputs are at differing logic levels (e.g., one HIGH, one LOW). If the inputs match, the output is LOW. This characteristic makes it indispensable for binary arithmetic addition, parity generation, and error detection. The 7486 IC contains four independent 2-input XOR gates.

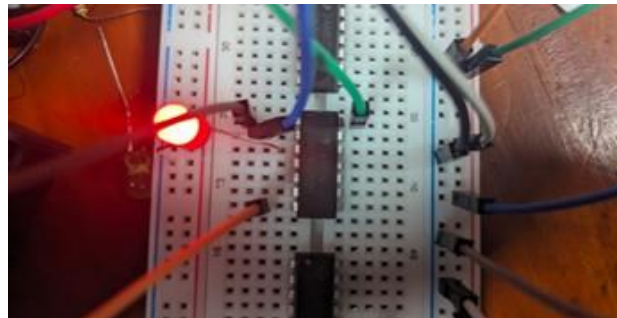


Fig 6: XOR Gate

7.XNOR Gate (IC 74266 / IC 4077): The Exclusive-NOR (XNOR) gate acts as an equivalence or equality detector. It produces a HIGH (1) output when its inputs are identical (both HIGH or both LOW), and a LOW (0) output when the inputs diverge. Acting as the precise inverse of the XOR gate, it is highly utilized in data verification and digital comparison systems.

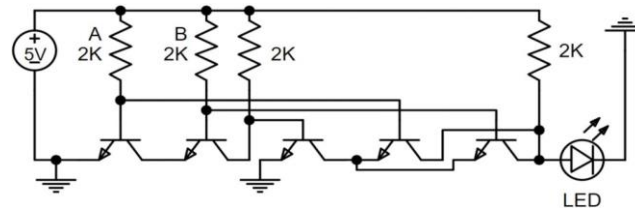


Fig 7: XNOR Gate

Discussion

Throughout this laboratory exercise, the core logic behaviors of AND, OR, NOT, NAND, NOR, XOR, and XNOR gates were successfully and empirically validated utilizing standard TTL integrated circuits. The implementation required meticulous attention to breadboard architecture, specifically regarding proper IC placement across the central trench and accurate pin-to-pin wiring. The standard 74-series ICs utilized feature durable outer plastic housings with standardized pin numbering, which greatly streamlined the identification and prototyping process.

By employing tactile push buttons to inject logic signals and LEDs to visually monitor the resulting output states, the physical hardware behavior was proven to align flawlessly with established theoretical Boolean truth tables. The accuracy of these results was highly dependent on maintaining a stable +5V DC power supply, ensuring correct VCC and GND biasing for every chip, and adhering strictly to the pin diagrams. Ultimately, this exercise underscored the reliability of TTL ICs as the foundational architecture for complex digital electronics and highlighted the necessity of systematic troubleshooting in hardware development.

Conclusion

This experiment successfully fulfilled all stated objectives by demonstrating the operational dynamics of fundamental digital logic gates through direct hardware implementation. The empirical verification of the truth tables for the AND, OR, NOT, NAND, NOR, XOR, and XNOR gates confirmed absolute alignment between theoretical Boolean logic and physical circuit behavior. Furthermore, the hands-on methodology cultivated essential practical engineering skills, including prototyping, component integration, and systematic troubleshooting, providing a robust conceptual and practical foundation for future studies in advanced digital system design.

References

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- [2] M. Morris Mano and M. D. Ciletti, *Digital Design*, 5th ed., Pearson Education, 20

